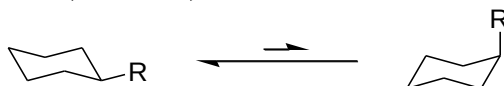


Lecture “Stereochemistry”

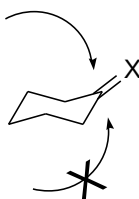
Take-home messages 2

3. Conformational analysis of cyclohexane derivatives

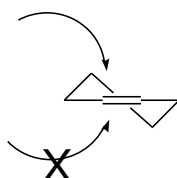
Cyclohexanes: - Two chair conformers, substituents better in equatorial position (A-values)



Cyclohexenes: - **exocyclic** double bond: axial attack of nucleophiles preferred



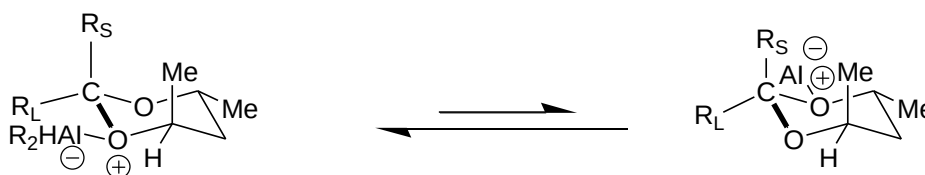
- **endocyclic** double bond: Fürst-Plattner-Rule (no twist conformation)



4. Conformational analysis of saturated heterocycles

Acetals: - Cyclic acetals show higher A-values due to shorter bond length
- Formation of oxonium species by Lewis acids

Chiral acetals: - Opening to the oxonium species occurs in way which removes the greatest amount of unfavourable steric repulsion



5. Cyclic Stereocontrol

Cyclopropanes/Cyclobutanes: - rigid, well defined upper and lower ring side

Cyclopentanes: - more flexible, sometimes surprises in prediction

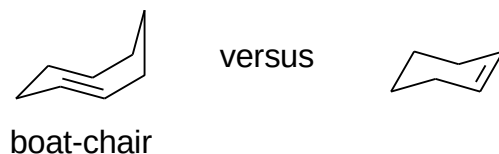
Cyclohexanes: - well defined, prediction possible according to A-values, Fürst-Plattner-Rule, etc.

Cycloheptane:

- flexible, typically difficult to predict conformation

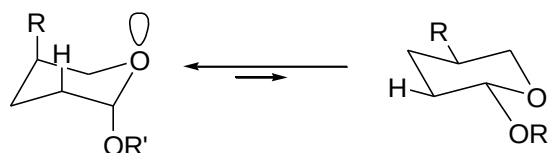
Cyclooctanes:

- boat-chair conformation, pseudo-A-values, in all larger rings double bonds have an inner and an outer side (perpendicular to ring plane)

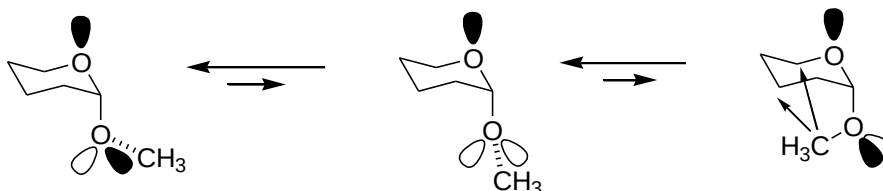
**6. Stereoelectronic effects**

Effects upon structure/reactivity by orbital interaction between σ/σ^* -orbitals and unfilled/filled orbitals on neighbouring centers.

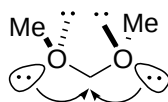
Endo anomeric effect: Electronegative substituents adjacent to oxygen prefer axial position



Exo anomeric effect: Rotation of OR group in anomeric position is “locked”

**Gauche effect:**

- Acyclic acetals prefer gauche conformation over linear conformation
- Rotational barrier in esters and amides, higher reactivity of lactones compared to esters

 **β -Silyl effect:**

- Silicon atoms stabilize carbocations at β -position

